SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

COMPUTER SCIENCE AND ENGINEERING PROGRAMME

CSA37 – SOFTWARE TESTING

EXPERIMENT 4

E-COMMERCE APPLICATION – POSITIVE TEST SCENARIOS AND TEST CASES

Aim:

To write a case study to identify and analyze the requirements of E-Commerce Applications, and prepare positive test scenarios as well as test cases for the application.

1. Problem Description

An e-commerce application enables customers to register, log in, search and filter products, view product details, add available products to a cart, provide a delivery address, pay online and track orders.

The E-Commerce Application is expected to provide reliable, secure and user-friendly services. The testing team verifies the stated requirements using valid and invalid inputs, records actual results, and reports any deviation from the expected behaviour.

2. Functional Requirements

1. Users must register with valid name, email, password and mobile number.
2. Users must log in using valid credentials.
3. Product search must return relevant products.
4. Only available products can be added to the cart.
5. Quantity must be greater than zero and not exceed stock.
6. The cart must calculate item totals, discounts, tax and final amount correctly.
7. Valid delivery details must be accepted.
8. An order is confirmed only after successful payment.

3. Positive Test Scenarios and Test Cases

Positive testing checks whether the application performs the required operation when valid data and correct user actions are supplied.

Test Case ID	Test Scenario	Test Input / Action	Expected Result	Status
PT01	Register new customer	Enter valid name, email, password and mobile number	Customer account is created.	Pass
PT02	Login	Registered email and correct password	Customer logs in successfully.	Pass
PT03	Search product	Search for “Laptop”	Relevant products are displayed.	Pass
PT04	Filter products	Select a valid brand and price range	Only matching products are displayed.	Pass
PT05	View product details	Open an available product	Price, description, images and stock status are shown.	Pass
PT06	Add product to cart	Add one in-stock product	Product appears in cart.	Pass
PT07	Update quantity	Change quantity from 1 to 2 when stock is sufficient	Quantity and total price are updated.	Pass
PT08	Place order	Enter valid address and complete successful payment	Order confirmation and order ID are displayed.	Pass

5. Detailed Execution of a Failed Test Case

Test Case ID: PT08

Precondition: A registered user has a valid account and the required service is available.

Test Steps: 1. Sign in with valid credentials. 2. Open the relevant module. 3. Enter valid data. 4. Submit the operation.

Expected Result: The application completes the operation and displays a success confirmation.

Actual Result: The operation completes successfully in this illustrative execution.

Conclusion: The positive test passes when actual behaviour matches the expected result.

Conclusion

The test cases prepared for the E-Commerce Application provide coverage of its key functional requirements. Executing them helps identify functional gaps, improve error handling and increase confidence in the quality of the application.